

hidden layer acts as a summary layer for the entire sentence.

The decoder, on the other hand, generates another series, one word at a time. The decoder is influenced by the context and by previously generated words.

Generally, this model is best suited to fixed length sequences; however, before applying the training to the model, padding concepts are used to convert variable length series into fixed length series. For example:

```
Query : [P P P P P P "What" "About"
"Placement" " ?" ]
// Assume that the fixed length is
10.P is Padding
Response : [ SD " It" "is" "Almost"
"100%" END P P P P ]
// SD means start decoding. END means
response is over. P is Padding
```

Word embedding is another important aspect of deep learning based chatbots. It captures the context of the word in the sentence, the semantic and syntactic similarities, as well as the relationship with other words. Word2Vec is a famous method to construct word embedding. There are two main techniques in Word2Vec and both are based on neural networks — continuous bag-of-words (CBOW) and continuous skip-gram.

The continuous bag-of-words method is generally used as a tool of feature generation. A sentence is first converted into a bag of words. After that, various measures are calculated to characterise the sentence. The frequency is the main measure of the CBOW. It provides better accuracy for frequent words. The skip-gram method achieves the reverse of the CBOW method. It tries to predict the source context words from the target. It works well for fewer training data sets.

The logic for the chatbots that use deep learning is as follows:

Step 1: Build the corpus vocabulary.

Step 2: Map a unique numeric identifier with each word.

Step 3: Padding is done to the

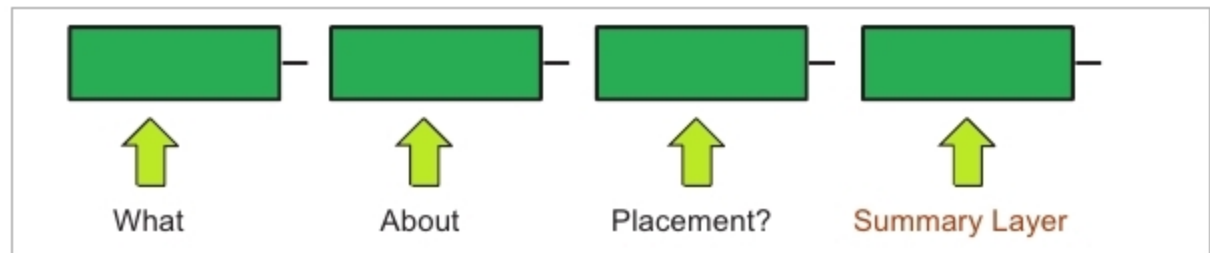


Figure 2: Encoder layers

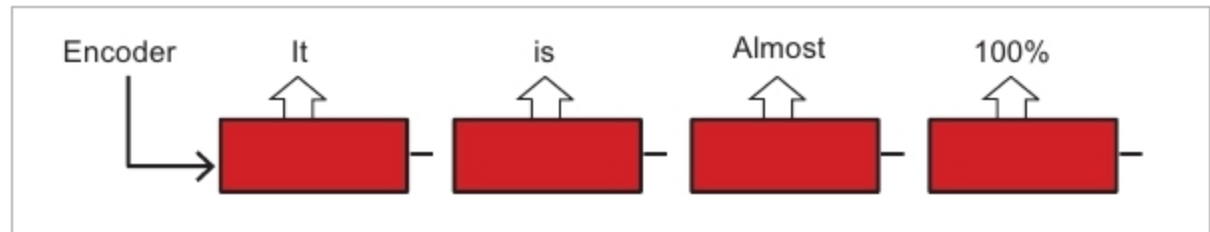


Figure 3: Decoder functioning

context words to keep their size fixed.

Step 4: Make a pair of target words and surround the context words.

Step 5: Build the deep learning architecture for the CBOW model. This involves the following sequence:

- Input as context words
- Initialised with random weights
- Arrange the word embeddings
- Create a dense softmax layer
- Predict target word
- Match with actual target word
- Compute the loss
- Perform back propagation to update embedding layer

Step 6: Train the model.

Step 7: Test the model.

Deep learning tools for chatbots

TensorFlow is a great tool that uses deep learning. It uses linear regression to achieve effective conversation. We first need to develop a TensorFlow model by using JSON to recognise patterns. The next step is loading this framework and contextualising the data. TensorFlow makes chatbots realistic and very effective.

Microsoft conversational AI tools are another important resource to design effective chatbots. These tools can be used to design, link, install and accomplish intelligent bots. The Microsoft Bot Builder software development kit (SDK) is ideal for the quick, free and easy development of chatbots with intelligence.

Pytorch is an excellent open source library based on Python for applications like chatbots. The Optim module implements various algorithms based on neural networks, which are essential for the designing of efficient chatbots. It also provides the power of Tensors and so has the same functionalities as TensorFlow.

Chatbots are essential if organisations aim to deal with customers without any human intervention. As discussed, deep learning based chatbots are the better option compared to the traditional variants, as the former handle abundant data efficiently. And generative models for building chatbots are more appropriate in the modern context. **END** 🐧

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